

Units and Dimensional Analysis Review

Units are quite useful when solving physics problems. If your answer has the wrong units, it is wrong! You may also be able to more-or-less solve your problem just using units. If your answer can only depend on a few quantities, and there's only one way to arrange those quantities to produce the desired units, then you have your answer. You can often produce seemingly miraculous results with almost no effort just by relying on units and dimensional analysis (the art of making sure your units are correct).

1. Velocity

Let's start with velocity/speed (since we don't actually have directions for these examples, velocity and speed are interchangeable in this section). Velocity is the rate of change of position with time. We can measure distance in metres, km, feet, angstroms, Planck lengths, miles. Time we measure in seconds, minutes, hours, days, years, Hubble times, etc. If I walk 10 km in 2 hours, my velocity is 10 km/2 hours or 5 km/hour. How far would we go if we walked for 4 hours? In this example, you could probably just say "20 km", but let's look in a bit more detail. We're looking for a distance (how far we walk). The units of distance we're using are km, so our final answer should have units of km. How do I go from km/hour to km? I had better multiply by some number of hours in order to cancel the hour in the velocity. Mathematically,

$$5 \frac{km}{hour} \times 4 \text{ hours} = 20 \frac{km \text{ hour}}{hour} = 20 \text{ km} \quad (1)$$

In the last step, because the hours appear in both the top and bottom, they cancel each other out leaving one. Just like $5/5 = 1$, so also $hour/hour = 1$.

If we had tried to do anything besides multiplying, we would not have gotten the hours to cancel, and our final answer would, by definition, be wrong. Say we had divided by the time instead. Then we would have gotten

$$5 \frac{km}{hour} / 4 \text{ hours} = 1.25 \frac{km}{hour^2} \quad (2)$$

We're after km , but we got $km/hour^2$. Those are not the same thing, so it is physically impossible for our answer to be correct. Always keep track of your units – it's a very powerful way of making sure your answer might be right. You can sometimes get a wrong answer with the right units, but you can never get a right answer with wrong units.

2. Unit Systems and Conversions

The units of velocity are distance divided by time, but we are in principle free to pick any measure of distance and any measure of time. In the previous example, we used km for distance and hours for time, but we could have picked anything. If I want to compare two velocities, though, it's a lot simpler if both velocities use the same measure of distance and time. Is 30 km/hour faster or slower than 10 m/s? It's not obvious at first glance. Is 30 km/hour faster or slower than 25

km/hour? That's much easier. To avoid confusion, we generally pick one measure of each basic unit and stick to it. In science, we usually use either MKS (for metres, kilograms, and seconds) or CGS (centimetres, grams, and seconds). In this class, we'll use MKS (also known as System International or SI), so every distance will be in metres, and every mass in kilograms, and every time in seconds.

Let's take the 5 km/hour we had from the last section and convert it to an MKS speed. We can always multiply an expression by one without changing it, and the same thing is true for units. Let's start by converting the km to metres. We know that 1 km=1000 m, and we want to replace the km with metres. We know that

$$1 \text{ km}/1000 \text{ m} = 1000 \text{ m}/1 \text{ km} = 1 \quad (3)$$

We want to get rid of the *km* in the numerator, so we should multiply by something with *km* in the denominator. That leaves us with

$$5 \frac{\text{km}}{\text{hour}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 5000 \frac{\text{km m}}{\text{hour km}} = 5000 \frac{\text{m}}{\text{hour}} \quad (4)$$

We've gotten the distance part sorted, but our time is still in hours. We repeat the exercise using the fact that 1 hour = 3600 seconds. This time, the hour is the denominator, so we want to multiply by a factor with hour in the numerator to get the hours to cancel. That gives us

$$5000 \frac{\text{m}}{\text{hour}} \times \frac{1 \text{ hour}}{3600 \text{ s}} = \frac{5000 \text{ m hour}}{3600 \text{ hour s}} = 1.39 \frac{\text{m}}{\text{s}} \quad (5)$$

By making sure the units worked out, we didn't have to guess at the right operation. There was only one way to get the units we wanted, and we could just use the units to reason what the right way to convert units was. At first this sort of operation can feel a little slow and awkward, but it rapidly becomes second nature with practice.

3. Acceleration

Much like velocity is the change in distance with time, acceleration is the change in velocity with time. Velocity is in *m/s* and we have to divide that by the time over which the velocity changed, so the units of acceleration are *m/s* per second, or *m/s²*. If we multiply two copies of the same unit together, the units just square in exactly the same way that $3 \times 3 = 3^2 = 9$.

For an example, let's look at the acceleration of cars. Before we work out numbers, it's good to have an idea of the sort of answer we expect to get. The most familiar acceleration we have is from Earth's gravity, which is around $g = 9.8 \text{ m/s}^2$. How does the car's acceleration compare? Qualitatively, if our acceleration matched the Earth's, we'd feel our weight double¹. I don't feel like I have someone sitting in my lap when a car accelerates, so the acceleration should be noticeably smaller than 9.8 m/s. If I hit the gas with an open cup of something, though, if the cup is anywhere

¹I'm technically assuming that the acceleration is in the same direction as gravity, but if I were more careful the answer wouldn't change very much.

close to full, the contents will spill all over the car. That happens because to the cup, “down” is the direction of the total acceleration (from the car plus Earth’s gravity). When we accelerate, “down” to the cup changes direction, and the liquid in the cup sloshes around. We wouldn’t get much sloshing if the direction of “down” didn’t change very much. Because we do get sloshing, our acceleration must be at least somewhat comparable to g . That leaves us expecting an answer where the car’s acceleration should be less than g , but not a whole lot less.

We can put in numbers now. A typical 0-60 time for a non-sports car is say 8 seconds. That means our speed goes from 0 to 60 miles per hour over 8 seconds. Our acceleration is then

$$60 \frac{\text{miles}}{\text{hour}} / 8 \text{ seconds} = 7.5 \frac{\text{miles}}{\text{hour} \times \text{second}} \quad (6)$$

Those units aren’t great - we have mixed English units with hours *and* seconds, to make a monstrosity of a unit. But wait! We learned how to convert units, so let’s convert these to MKS. We want to start by turning miles in metres. There are 1.6 (approximately) km in a mile, so we can convert miles to metres like this:

$$1 \text{ mile} = 1 \text{ mile} \times \frac{1.6 \text{ km}}{1 \text{ mile}} = 1 \text{ mile} \times \frac{1.6 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} \quad (7)$$

If we apply Equation 7 to hour acceleration, we get

$$7.5 \frac{\text{miles}}{\text{hour} \times \text{second}} = 7.5 \frac{\text{miles}}{\text{hour} \times \text{second}} \times \frac{1.6 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 12,000 \frac{\text{m}}{\text{hour} \times \text{s}} \quad (8)$$

We aren’t out of the woods yet because we have one copy of hours in the denominator. We want to convert that to seconds as we did in the velocity example:

$$12,000 \frac{\text{m}}{\text{hour} \times \text{s}} \times 1 \text{ hour} / 3600 \text{ s} = \frac{12,000 \text{ m}}{3600 \text{ s}^2} = 3.3 \frac{\text{m}}{\text{s}^2} \quad (9)$$

This is about a third of Earth’s gravity, and is a perfectly reasonable number. We expected something smaller than g , but not a whole lot smaller. That’s exactly what we ended up with.

4. Some Other Units

Let’s look at some other units that show up. From Newton’s second law, we know that

$$F = Ma \quad (10)$$

where F is the force, M is the mass², and a is the acceleration. Mass is in kg and acceleration is in m/s^2 , so the units of force must be

$$Ma = kg \times \frac{m}{s^2} = \frac{kg \text{ m}}{s^2} \quad (11)$$

²Although Newton’s second law is usually written $F = ma$, I’m using the capital M to avoid confusion with the unit m for metre.

In honor of Sir Isaac who came up with the second law, we call this unit of force the *newton* or N . Even though it comes from a person's name, we do not capitalize newton when it's used as a unit.

$$1\ N = 1\ \frac{kg\ m}{s^2} \quad (12)$$

You may recall that work equals force time distance. The units of work must then be

$$N\ m = \frac{kg\ m}{s^2} \times m = \frac{kg\ m^2}{s^2} \quad (13)$$

This is a *joule* or J , named for James Prescott Joule. Joule the person was a brewer who worked out the relation between heat and work in his spare time. One joule is the amount of energy it takes to move 1 metre against a force of 1 N.

Pressure is a force per area, so its units must be

$$N/m^2 = N\ m^{-2} \quad (14)$$

This is the *pascal* of P , and air pressure at sea level is very close to $10^5\ P$. Now we can do something kind of fun (depending on your definition of fun). We take a pascal, and multiply by $1\ m^3 / m^3$ which gives us

$$1\ P = 1\ \frac{N}{m^2} \times \frac{1\ m^3}{1\ m^3} = 1\ \frac{N\ m}{m^3} = 1\ \frac{J}{m^3} \quad (15)$$

A joule per cubic metre is an energy *density*.

Because they have the same units, we might expect pressure to be very closely related to energy density. Indeed they are related, and for an ideal gas the energy density U is

$$U = \frac{3}{2}nkT \quad (16)$$

where n is the number density of gas molecules, k is the Boltzmann constant, and T is the temperature, in Kelvin. From the ideal gas law, we know that $P = \frac{N}{V}kT$ so we know that

$$U = \frac{3}{2}nkT = \frac{3}{2}P \quad (17)$$

This is an example of dimensional analysis and order-of-magnitude estimation. It takes actual detailed physics knowledge of statistical mechanics to get the factor of 1.5 correct. By looking at units, though, we could get a fairly accurate guess without doing any hard physics at all! We use similar estimates in physics all the time. If I'm faced with a new problem that I have no idea how to solve, I write down every quantity that might affect the answer, and figure out how to combine those quantities to give me the right units. If there's only one way to do it (and there had better be at least one way; otherwise you can't make an answer with correct units, and answers with incorrect units are wrong!), then I have a good guess for what my answer should be. There will (usually) be some factor that we'll miss this way, but those factors are almost always "close-to" one.

5. Atomic Tests

To show the power of dimensional analysis, let's look at one of the most famous examples: the Trinity nuclear test. Trinity was the first nuclear explosion, and *Life Magazine* published a picture of it. The Soviets were obviously *very* interested in how powerful the US's new weapon was, and the *Life* photo let them figure it out. The photo includes a time stamp and a scale bar, so you know how big the explosion was at a given point in time. That is enough info to make a really good estimate of the energy released by the nuclear bomb.

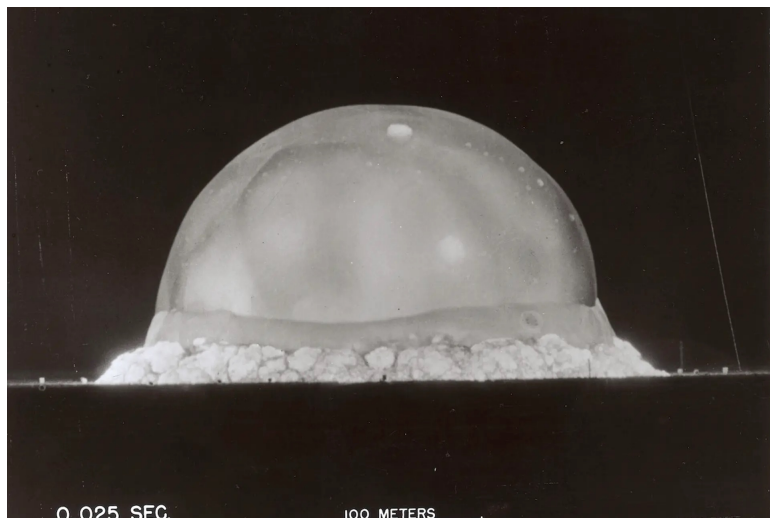


Fig. 1.— Trinity nuclear test. Note that both a time and length scale were given in the photo, which enabled solving for the energy from dimensional analysis.

We want to know how much energy the bomb released. We have a time and a size. What else could it depend on? The explosion is going into the atmosphere, so the density of air probably plays a role. No air, and the explosion moves a lot faster. How about air pressure? That might play a role, but if we think about it for a second, we can convince ourselves that air pressure probably does not matter. If you look at the photo in Figure 1, the blast is 100 metres across at 0.025 seconds, so the typical speed is $100 \text{ m}/0.025 \text{ s} = 4000 \text{ m/s}$. That is a lot larger than the speed of sound, which means air pressure is tiny compared to the pressure from the explosion. That leaves us with a length (metres), time (seconds) and density (kg per cubic metre) and we want to make an energy (kgm^2/s^2). We only have one kg to work with, so the energy had better be proportional to the density. The density ρ gives us a m^{-3} , but we need to get to m^2 , so we'll need to multiply by distance d to the fifth power. Finally, we also need to get s^{-2} in our energy, so we had better divide by the time t squared. We now have the combination of quantities that gives us the right units:

$$E = \frac{\rho d^5}{t^2} \quad (18)$$

We can now plug in numbers to estimate the energy. The radius of the blast in the picture is say 150 m and 0.025 s , and the density of air is around 1 kg/m^3 . Plug everything in and we have

$$E = \frac{1 \cdot 150^5}{0.025^2} \text{ J} = 1.2 \times 10^{14} \text{ J} \quad (19)$$

That's about 30 kilotons of TNT³. That's really quite close to the actual answer, 25 kton. Of course, our answer could have been off somewhat - doing the actual calculation is hard! But dimensional analysis let us (and the Soviets) a surprisingly accurate answer without having to do detailed physics modelling.

Don't worry if you didn't follow all the details of the Trinity calculation. The point was that we could get a surprisingly accurate answer to a very, very hard question just by making sure our units worked out!

³The energy in TNT varies, but for simplicity people have adopted 1 gram of TNT releases 1 kilocalorie of energy. That's actually not very much - proteins and carbs have 5-6 kcal/g, and fats have 9 kcal/g. The secret to explosives is not always how much energy they release, but how quickly they release it.